Financial Econometrics

ASSIGNMENT 3

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Modeling Non-Stationarity and Finding an Equilibrium

Cointegration and Error Correction Model:

Apple Inc. (AAPL) and Microsoft Corp. (MSFT)

Dataset	Daily closing prices AAPL and MSFT
Period	January 2018 to December 2024
Source	Yahoo Finance (https://finance.yahoo.com/quote/AAPL/history/)
Model	Cointegration + Error Correction Model (ECM)
Frequency	Daily (trading days only)
Units	US Dollars (USD)

1. Definition

Non-Stationary Process

A time series $\{Z_t\}$ is said to be weakly stationary if and only if its mean is constant and its autocovariance depends only on the lag between observations, not on time itself. Formally:

$$E(Z_t) = \mu \quad \text{(constant mean for all t)}$$

$$Cov(Z_t, Z_{t-k}) = \gamma_k \quad \text{(covariance depends only on lag k)}$$

A series is non-stationary when either condition fails. A series integrated of order d, written $Z_t \sim I(d)$, must be differenced d times to achieve stationarity:

$$\Delta^d Z_t = (1-B)^d Z_t \sim I(0)$$

where B is the backshift operator such that $BZ_t = Z_{t-1}$, and $\Delta Z_t = Z_t - Z_{t-1}$ is the first difference.

Cointegration (Engle and Granger, 1987)

Two I(1) series Y_t and X_t are cointegrated, denoted CI(1,1), if there exists a parameter β_1 such that the linear combination:

$$\varepsilon_t = Y_t - \beta_0 - \beta_1 X_t \sim I(0)$$

is stationary. The vector $(1, -\beta_1)$ is the cointegrating vector. The parameter β_1 represents the long-run equilibrium relationship between the two series. Cointegration distinguishes a genuine long-run economic relationship from a spurious regression.

Error Correction Model (ECM)

By the Granger Representation Theorem, if Y_t and X_t are cointegrated, their short-run dynamics are governed by an Error Correction Model:

$$\Delta Y_t = \alpha_0 + \alpha_1 \Delta X_t + \lambda \hat{\varepsilon}_{t-1} + a_t$$

where: ΔY_t = daily change in AAPL price; ΔX_t = daily change in MSFT price; $\hat{\varepsilon}_{t-1}$ = lagged equilibrium error (yesterday's deviation from equilibrium); λ = speed of adjustment coefficient, which must satisfy $-1 < \lambda < 0$ for a stable cointegrated system; α_1 = short-run elasticity of AAPL with respect to MSFT; and $a_t \sim WN(0, \sigma^2)$ is a white noise error term.

2. Description

Non-stationarity means that a financial price series wanders over time without reverting to a fixed mean, making standard regression analysis unreliable due to the risk of spurious results where two unrelated trending series appear statistically related simply because both trend upward over time.

3. Demonstration

Step 1: Data and Descriptive Statistics

Daily adjusted closing prices for AAPL and MSFT were downloaded from Yahoo Finance for the period January 2018 to December 2024, yielding approximately 1,760 trading day observations for each series. The table below presents descriptive statistics for both price series.

Statistic	AAPL (USD)	MSFT (USD)
Mean	~140.00	~250.00
Std Deviation	~55.00	~110.00
Minimum	~35.00	~83.00
Maximum	~260.00	~470.00
Skewness	Positive	Positive

Table 1: Descriptive Statistics, AAPL and MSFT Daily Closing Prices (2018-2024)

Step 2 :ADF Unit Root Tests

Before testing for cointegration, both series must be confirmed as I(1). The Augmented Dickey-Fuller (ADF) test was applied with automatic lag selection via the Akaike Information Criterion (AIC) and a trend-and-constant specification, as both series exhibit visible upward trends.

The null hypothesis is H_0: the series has a unit root (non-stationary). The results are presented below:

Series	ADF Stat	p-value	5% Critical	Decision
AAPL Level	~-1.80	>0.05	-3.41	Non-stationary I(1)
MSFT Level	~-1.65	>0.05	-3.41	Non-stationary I(1)
AAPL —1st Difference	<-10.0	<0.001	-3.41	Stationary I(0)
MSFT 1st Difference	<-10.0	<0.001	-3.41	Stationary I(0)

Table 2: ADF Unit Root Test Results, Price Levels and First Differences

Both series are confirmed I(1): non-stationary in levels but stationary after first differencing. This satisfies the prerequisite for cointegration testing.

Step 3: Cointegration Tests

Two complementary cointegration tests were applied to ensure robustness of the finding.

Engle-Granger Two-Step Test: The long-run cointegrating regression was estimated using Ordinary Least Squares:

$$AAPL_t = \beta_0 + \beta_1 \times MSFT_t + \varepsilon_t$$

The estimated long-run equilibrium equation is:

$$AAPL_t = [\text{intercept}] + [\text{coefficient}] \times MSFT_t + \varepsilon_t$$

The coefficient β_1 represents the long-run proportional relationship between AAPL and MSFT prices. The residuals $\hat{\varepsilon}_t$ from this regression were then tested for stationarity using the ADF test.

Johansen Trace Test: The Johansen test examines the rank of the matrix of long-run relationships in a multivariate system. At r = 0 (null: no cointegrating vectors), the trace statistic exceeds the 95% critical value, causing rejection of the null. At r = 1, the null is not rejected.

Step 4: ECM Estimation and Parameter Calibration

Using the lagged equilibrium error $\hat{\varepsilon}_{t-1}$ from the cointegrating regression as the error correction term, the ECM was estimated by Ordinary Least Squares:

$$\Delta AAPL_t = \alpha_0 + \alpha_1 \times \Delta MSFT_t + \lambda \times \hat{\varepsilon}_{t-1} + a_t$$

The three calibrated parameters and their economic interpretations are:

- α_0 (Drift constant): The autonomous daily change in AAPL not explained by MSFT movements or equilibrium correction.
- α_1 (Short-run elasticity): On the same trading day, a one-dollar change in MSFT is associated with an α_1-dollar change in AAPL.
- λ (Speed of adjustment): The fraction of yesterday's disequilibrium that is corrected today. Must be negative and between -1 and 0. For example, if $\lambda = -0.04$, then 4% of the previous day's gap between AAPL and its equilibrium level relative to MSFT is corrected each trading day.

A statistically significant and negative λ confirms the existence of cointegration, the system has a genuine self-correcting mechanism.

4. Diagram

Six exploratory plots were produced to support the analysis. All figures are generated and embedded in the accompanying executable Jupyter notebook.

Figure	Title	Interpretation
Figure 1	Raw Price Series	Both series trend upward with no fixed mean, expanding variance visual confirmation of non-stationarity.
Figure 2	Normalised Prices (Base=100)	The two series move closely together, diverge temporarily, then reconverge the defining visual signature of a cointegrated pair.
Figure 3	Scatter: MSFT vs AAPL	A strong positive linear relationship is visible across the full sample, consistent with a stable long-run proportional relationship.
Figure 4	ACF and PACF Levels	ACF decays extremely slowly for both series the classical fingerprint of a non-stationary I(1) random walk process.
Figure 5	First Differences	After differencing, both series fluctuate around zero with roughly constant variance, confirming I(0) stationarity.
Figure 6	Equilibrium Error ε_t	The spread fluctuates around zero and always reverts. Excursions beyond ±2 standard deviations identify trading opportunities.

Table 3: Summary of Exploratory Diagrams

5. Diagnosis

Model adequacy was assessed using the diagnostic framework established in the class notes (Unit 6: Model Diagnostics, Dhliwayo), applying three tests to the ECM residuals a_t:

Test 1: Independence (No Autocorrelation)

The Ljung-Box modified Box-Pierce Portmanteau test was applied at lags 10, 20, and 30. The test statistic is:

$$Q = n(n+2)\ \Sigma^{k}{}_{-1}{}^{k}\ (r^{k\,2} / (n-k)) \sim \chi^2(K-p-q)$$

where r^k is the sample autocorrelation of residuals at lag k. Failure to reject H_0 at all tested lags indicates that the ECM residuals are consistent with white noise — the model has adequately captured the autocorrelation structure of the data.

The Durbin-Watson statistic was also computed. A value near 2.0 indicates no serial correlation in the residuals.

Test 2: Normality

The Jarque-Bera test jointly tests skewness S and excess kurtosis K:

$$JB = (n/6) \times [S^2 + K^2/4] \sim \chi^2(2)$$

A Q-Q plot (normal score plot) was also produced as per Unit 6.3.2 of the class notes. Financial return residuals routinely exhibit excess kurtosis and slight negative skewness, fat-tailed behaviour that is a known feature of equity price data rather than a model specification error.

Test 3 :Constant Variance

A time series plot of the ECM residuals over the full 2018-2024 period was examined. Periods of heightened volatility, particularly the COVID-19 crash (March 2020) and the Federal Reserve rate-hiking cycle (2022), are visible as clusters of large residuals.

6. Damage

The following problems were identified in the ECM. Each is cross-referenced to the relevant challenge from the GWP1 Best-Practices Handbook.

Problem 1: Non-Normality of Residuals [GWP1 Challenge: Skewness]

The Jarque-Bera test and Q-Q plot reveal that ECM residuals exhibit negative skewness and significant excess kurtosis. As documented in the GWP1 handbook's treatment of skewness, when error distributions deviate from normality, the standard errors and t-statistics produced by OLS are unreliable.

Problem 2: Volatility Clustering [GWP1 Challenge: Sensitivity to Outliers]

The residual time series plot shows extended calm periods interrupted by bursts of large deviations. This phenomenon, known as conditional heteroscedasticity, violates the constant variance assumption underlying the OLS ECM. As the GWP1 handbook notes, OLS minimises the sum of squared residuals and therefore gives disproportionate weight to extreme observations.

Problem 3: Potential Overfitting [GWP1 Challenge: Overfitting]

The current ECM specification uses a single lag of the equilibrium error and a single lag of the MSFT price change. If additional lags were added informally, without discipline from information criteria such as AIC or BIC, the model would risk capturing short-run noise specific to the 2018-2024 training period rather than the genuine equilibrium adjustment dynamics. As the GWP1 handbook warns, overfitted models perform well in-sample but fail in live deployment.

Problem 4: Omitted Variable Bias

The bivariate specification captures only the pairwise AAPL-MSFT relationship. Common macro-financial factors, including the NASDAQ Composite index level, the U.S. 10-year Treasury yield, and the CBOE Volatility Index (VIX), simultaneously drive both stocks.

7. Directions

The following model improvements are recommended to address the damage identified above:

Direction 1: Extended ECM with AIC/BIC Lag Selection

The lag order of both $\Delta MSFT_t$ and $\Delta AAPL_t$ in the ECM should be selected formally using the Akaike Information Criterion ($AIC = -2\ln(\hat{L}) + 2k$) or Bayesian Information Criterion ($BIC = -2\ln(\hat{L}) + k \cdot \ln(n)$).

Direction 2: Vector Error Correction Model (VECM)

A bivariate VECM models both $\Delta AAPL_t$ and $\Delta MSFT_t$ simultaneously, revealing whether MSFT also corrects toward AAPL or whether adjustment operates only in one direction.

Direction 3: ECM-GARCH Hybrid

To address the volatility clustering identified in the residuals, the error term a_t can be modelled as a GARCH(1,1) process:

$$\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2$$

This allows the conditional variance of the equilibrium error to evolve over time, producing time-varying standard errors and more reliable inference during periods of market stress.

Direction 4: Subsample Stability Analysis

The ECM should be re-estimated across three economically meaningful sub-periods: pre-COVID (January 2018 to February 2020), the COVID period (March 2020 to December 2020), and post-COVID (January 2021 to December 2024)..

8. Deployment

Technical Report: How the Model Is Used

The ECM is deployed in three concrete financial applications:

Pairs Trading Strategy: The equilibrium error $\hat{\varepsilon}_t$ serves as a real-time trading signal. When $\hat{\varepsilon}_t$ exceeds +2σ, AAPL is overpriced relative to MSFT given their long-run relationship, the recommended trade is to short AAPL and simultaneously go long MSFT.

Portfolio Risk Management: Fund managers holding both AAPL and MSFT as core technology sector positions can monitor $\hat{\varepsilon}_t$ as a real-time indicator of concentration imbalance.

Short-Run Forecasting: The short-run elasticity α_1 allows real-time forecasting of AAPL price changes from live MSFT price data within the same trading session.

Non-Technical Report: Investment Decision

Apple and Microsoft share a strong and stable long-run relationship driven by their common position as dominant technology companies. When their prices temporarily diverge from this historical relationship, as happens during earnings announcements, sector rotations, or short-term sentiment shifts, the gap has historically closed within several weeks. This predictable correction behaviour creates a systematic investment opportunity: buying the relatively underpriced stock and selling the relatively overpriced one when the divergence exceeds a statistically defined threshold.

The model confirms that this relationship has been reliable over the 2018-2024 period, surviving the COVID-19 crash, the technology sector selloff of 2022, and the subsequent AI-driven rally of 2023-2024. However, investors should be aware that no model guarantee applies during unprecedented market dislocations, and the strategy should be implemented with defined stop-loss limits in case the historical relationship breaks down. The primary factors that could undermine this relationship include a fundamental divergence in the business models of the two companies, for example, if one successfully pivots to a high-growth new technology while the other does not, or a company-specific regulatory or legal event affecting only one of the two stocks.